

Attachment 4 – Geospatial Information and Services (GI&S) Deliverables

1. **Geospatial Data¹:** The Contractor shall deliver the GI&S data in an ESRI file geodatabase in accordance with the latest Air Force Adaptation version of Spatial Data Standards for Facilities Infrastructure and Environment (SDSFIE) for a seamless import of data into the appropriate HQ, MAJCOM, and/or Installation GeoBase Enterprise database. All Map Exchange Documents (MXDs) developed and/or used for the project shall be delivered with the aforementioned file geodatabases. MXDs shall have relative pathnames enabled within the map settings for each .MXD. All text and graphic data related to GI&S map exhibits shall be provided in MS Word, MS Excel and Adobe .pdf format. Deliverables intended for interaction with other databases or GeoBase feature classes or “layers” (i.e. file geodatabases, etc.) shall comply with the Open GIS Consortium (OGC) guidelines for interoperability on multiple systems.
 - Data shall conform to the latest Air Force Adaptation of SDSFIE-V as applicable, and SDSFIE-V Gold if no Component HQ Adaptation exists.
 - All GI&S data delivered will be in:
 - Projection: Applicable UTM Zone
 - Datum: WGS 1984
 - Geoid: EGM 96
 - Units: Meters
 - Information regarding latest Air Force Adaptation of SDSFIE can be found at: <https://www.sdsfieonline.org/Components/USAF>
 - For those data layers that were created for or during the performance of this project, all layers shall have at a minimum the latest version of SDSFIE Gold foundational attribute fields included and populated, as enumerated in the following table:

Table: SDSFIE Gold 4.x Foundational Attribution

SDSFIE Gold Attributes Attribute Name	Definition, Description and Note	Data Type	Nullable	Mandatory	Attribute Guidance
featureDescription	A narrative describing the feature	String (Max)	Yes	Yes	
featureName	The common name of the feature	String (80)	Yes	Yes	The official name of the feature found on installation maps or in authoritative databases of record.
installationId	The official code assigned by the Military Service	String (11)	Yes	No	The installation ID should match the Installation Code

¹ Shall encompass other data types (such as Google .KML) in addition to AutoCAD .dwg created/utilized during the project to be delivered in the ESRI file geodatabase format

SDSFIE Gold Attributes Attribute Name	Definition, Description and Note	Data Type	Nullable	Mandatory	Attribute Guidance
	(includes Washington Headquarters Services) used to identify the site or group of sites that make up an installation.				reported to the RPI. All sites and assets within an installation shall have the same installationId.
mediaId	Used to link the record to associated multimedia records that reference data Example media types include imagery, video, audio, scanned documents, and drawings. See your DoD Component service implementation guidance for details as to the content of this attribute.	String (40)	Yes	Yes	
metadataId	Used to represent or link to feature level metadata.	String (80)	Yes	No	
featureIdpk	Primary Key. A unique, user-defined identifier for each record or instance of an entity.	String (40)	No	Yes	
sdsId	A unique identifier for all entities in the SDSFIE	GUID	Yes	Yes	

- In addition to those fields enumerated in the SDSFIE Gold foundational attribution table, the inclusion of an Installation Name field is desirable for those datasets that do not have a corresponding AF SDSFIE Adaptation Data Content Specification, as appropriate:

installationName	The actual name of the installation that is associated with the installation ID defined by real property.	String (16)	Yes	No
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2. **CAD data:** CAD data shall be delivered in AutoCAD .dwg format. Units are to be set to meters with drawings drawn 1:1 in model space. CAD drawings are to be spatially correct, meaning they will correctly import to GIS without a need to scale or reorient the linework.
 - No .MXD documents shall contain reference to .dwg data formats. For those CAD datasets that are critical to the project and require mapping representation, .dwg files shall be converted to

file geodatabase format, comply with minimal foundational attribution (as enumerated in Table X, above), shall contain metadata and shall be packaged along with all other geospatial deliverables for this project.

3. **Metadata:** All geospatial submittals must contain metadata.

- It is preferable to use the Spatial Data Standards for Facilities, Infrastructure and the Environment – Metadata (SDSFIE-M) Metadata Style to populate metadata records. The SDFSIE-M Metadata Style for ArcGIS all available to download from: <https://www.sdsfieonline.org/Standards/Metadata>
- To populate metadata using the SDFSIE-M Metadata Style follow the directions below:
 - a. Click the download link and download the .zip file.
 - b. Extract the .zip file
 - c. Navigate into the resulting directory and into the *DISK1* directory
 - d. Double-click on *SDSFIE-M Metadata Style for ArcGIS 10.x.msi* and follow the instructions.
 - e. Once installed, you need to start ArcCatalog.
 - f. Navigate to the "Customize"->"ArcCatalog Options..." menu and select.
 - g. Click on the "Metadata" tab.
 - h. You should see the "Metadata Style" pulldown menu and should find "SDSFIE-M 2.0" to be one of the selections; select it.
 - i. Now, pick a vector data file and select the "Description" tab.
 - j. You should see a bunch of XML under the title: "Simple ArcGIS Metadata 1.0 Display"
 - k. If you click on the "Edit" button, you can begin editing the metadata for the selected data file.
 - l. Be sure to "Save" your changes.
 - m. Once you have cleared all of the Red X's, you can "Save" the SDFSIE-M (SMIS) record for the data file. A metadata record is considered compliant with SDFSIE-M 2.0 once all Red Xs are cleared using the SDFSIE-M 2.0 Metadata Style for ArcGIS.
- **If the SDFSIE-M Metadata Style is unavailable, metadata may also be populated using the ISO 19139 Metadata Implementation Specification Style.**

If populating metadata using the ISO 19139 Metadata Implementation Specification Style in ArcGIS, the "MD_Progress Code" (Resource >> Details >> Status) and "Maintenance" elements will need to be populated along with all ISO 19139 elements to ensure all SDFSIE-M 2.0 elements are included.

4. **Delivery Medium:** Geospatial deliverables shall be furnished to the appropriate government representative during the course of this contract to include pre-final and final stages.

- Pre-final Data may be transmitted by secure FTP using DoD SAFE (<https://safe.apps.mil>).
- DVDs containing final GI&S deliverables shall be provided as required, with number of copies to be determined by government representative

5. **Geospatial Vector check/Review:** Government representative has the right to make enquiry and receive information, to include copies of in-progress geospatial data and products, during the course of this contract.
- **At a minimum, government representative shall be furnished a copy of GI&S data/products created for and/or during the course of the project at the Pre-Final project stage. Government representative will strive to make comments and provide feedback within two (2) working weeks of receipt of Pre-Final GI&S deliverables.**
 - The Government shall review the submitted data and documentation upon completion of all stated work (Final deliverable stage). Missing or incomplete items will be documented and forwarded to the contractor for completion. Upon receipt of a complete submittal, the Government will conduct a quality review and notify the contractor within 30 days of acceptance (along with any stipulations this includes) or rejection of the deliverables described herein. Failure to adhere to any of the stated delivery specifications may result in rejection of deliverables and nonpayment.
6. **Source Data:** Source data and product data remain the property of the United States Government. The contractor may be required to explain and demonstrate the company's process for protecting all geospatial data, including but not limited to geometry, attributes, metadata, topologies, and relational database schemas and operations used in association with this SOW. The Contractor may be required to sign a non-disclosure agreement attesting to the same before source data is released. Further information about security and nondisclosure requirements should be obtained from the installation GIO. Some installation map data, source and/or product, may be considered by the Government to be "sensitive, but unclassified." The intent of this clause is to prevent intentional or unintentional dissemination of "sensitive, but unclassified" information to include unauthorized access to the source and product data by any entity wishing to do harm to the USAF or United States Government while the data resides on the Contractor's computer network. The contractor is not authorized to release this information to any third party without the explicit consent of Headquarters Air Force Office of the Civil Engineer (AF/A4C) or its involved Installations.